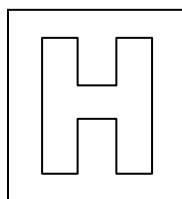


Candidate Name: _____

Class Adm No

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2011 Promotional Examination II Pre-university 2

H1 CHEMISTRY

8872 / 02

Paper 2

15 September 2011

2 hours

Additional Materials: Cover page
 Data Booklet
 Writing paper

READ THESE INSTRUCTIONS FIRST

Do not open this booklet until you are told to do so.

Write your name, index number and class on all the work you hand in.

Write in dark blue or black pen on both sides of the writing paper.

You may use a soft pencil for any diagrams, graphs or rough working.

Do not use staples, paper clips, highlighters, glue or correction fluids.

Section A

Answer **all** questions in the spaces provided.

Section B

Answer any **two** questions on the writing papers. Start each new question on a fresh sheet of writing paper.

Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of M_r and A_r , unless a different level of accuracy is specified in the question.

You are reminded of the need for clear presentation in your answers and to show all working in calculations.

The number of marks is given in brackets [] at the end of each question or part question.

FOR EXAMINER'S USE

	Section A			Section B			Total
Question No	1 (15m)	2 (10m)	3 (15m)	4 (20m)	5 (20m)	6 (20m)	80
Marks Obtained							

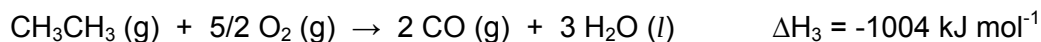
This question paper consists of 15 printed pages.

[Turn over

SECTION A (40 marks)

Answer all questions in the spaces provided.

- 1** The main compounds discussed in this question are ethane and ethene.
- (a)** When burnt in a limited supply of air, ethane forms carbon monoxide and water according to the reaction shown below.

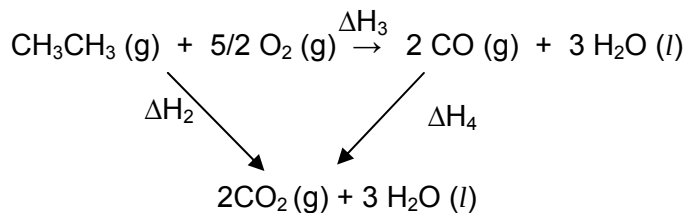


Ethane can be synthesized through the hydrogenation of ethene as shown in the following equation: $\text{CH}_2=\text{CH}_2 (\text{g}) + \text{H}_2 (\text{g}) \rightarrow \text{CH}_3\text{CH}_3 (\text{g}) \quad \text{-----} \Delta H_1$

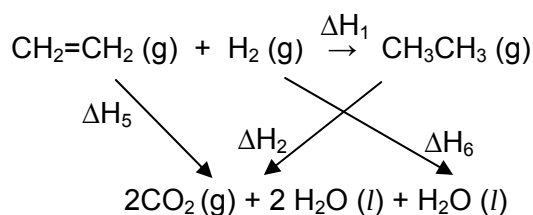
Given the following energy cycles and various enthalpy change data,

$\text{CH}_3\text{CH}_3 (\text{g}) + 5/2 \text{O}_2 (\text{g}) \rightarrow 2 \text{CO} (\text{g}) + 3 \text{H}_2\text{O} (\text{l})$	$\Delta H_3 = -1004 \text{ kJ mol}^{-1}$
Enthalpy change of combustion of $\text{CH}_2=\text{CH}_2 (\text{g})$	$\Delta H = -1411 \text{ kJ mol}^{-1}$
Enthalpy change of formation of $\text{H}_2\text{O} (\text{l})$	$\Delta H = -286 \text{ kJ mol}^{-1}$
Enthalpy change of combustion of $\text{CO} (\text{g})$	$\Delta H = -283 \text{ kJ mol}^{-1}$

Energy Cycle 1



Energy Cycle 2



- (i)** Identify the enthalpy changes represented by ΔH_4 , ΔH_5 , and ΔH_6 .

ΔH_4 :

ΔH_5 :

ΔH_6 :

- (ii) Use both energy cycles and the enthalpy change data given to calculate the enthalpy change of hydrogenation of ethene, ΔH_1 .

- (iii) Define the term *Standard Enthalpy Change of Combustion*, ΔH_c^θ .

.....
.....

- (iv) The carbon-oxygen bond energy of carbon monoxide is + 1074 kJ mol⁻¹ while the carbon-oxygen bond energy of carbon dioxide is + 802.5 kJ mol⁻¹. Why is the carbon-oxygen bond energy in carbon monoxide different from that in carbon dioxide?

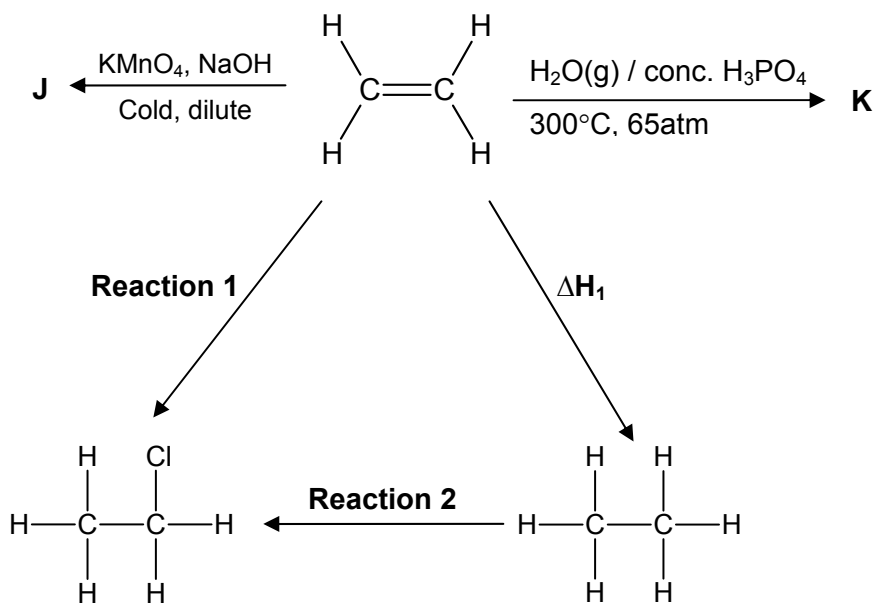
.....
.....

- (v) State one environmental consequence of excessive production of carbon monoxide arising from the combustion of car engines.

.....

[8]

(b) Consider the following reaction scheme:



(i) Draw the structure of the organic products **J** and **K**.

J	K

(ii) State suitable reagents and conditions required for:

Reaction 1

Reagents:

Conditions:

Reaction 2

Reagents:

Conditions:

[4]

- (c) Group I and Group II metals when burnt in air produces metal oxides.

The melting points of two such oxides are given below.

	Na ₂ O	MgO
melting point/ °C	1132	2852

- (i) Explain the difference in their melting points.

.....

.....

.....

.....

.....

.....

- (ii) Write an equation to describe how **any one** of these oxides reacts with dilute hydrochloric acid.

.....

[3]

[Total: 15]

2

Elemental nitrogen is a colorless, odorless and mostly inert diatomic gas at standard conditions, constituting 78.08% by volume of the Earth's atmosphere. Nitrogen has the symbol **N**, atomic number 7 and atomic mass 14.00674. Natural nitrogen consists of two stable isotopes, ^{14}N , which makes up the vast majority of naturally occurring nitrogen, and ^{15}N .

Nitrogen readily forms oxides of nitrogen when elemental nitrogen is combusted at high temperatures. Examples of oxides of nitrogen include N_2O_3 , N_2O_4 , and N_2O_5 , N_2O , NO_2 , and NO .

N_2O_3 , N_2O_4 , and N_2O_5 decompose rapidly at room temperature, N_2O is stable and rather unreactive at room temperature, while NO and NO_2 are quite reactive but stable when isolated.

Out of all these nitrogen containing compounds, N_2O_4 is one of the most important rocket propellants ever developed as it burns on contact without a separate ignition source, making it a powerful oxidizer.

- (a) Calculate the percentage abundance of the ^{15}N isotope.

[2]

- (b) State the electronic configurations of the elements N and O.

(i)

.....
.....

- (ii) Using the electronic configuration written in (b)(i), explain if the first ionization energy of N is higher or lower than the first ionization energy of O.

.....
.....
.....

[2]

- (c) By drawing suitable 'dot-and-cross' diagrams, explain the following statement in italics,
(i) '*N₂O is stable and rather unreactive at room temperature, while NO and NO₂ are quite reactive*'.

(*Hint: NO and NO₂ have structures that do not follow the octet rule)

.....
.....

- (ii) Compare the boiling points of N₂ and NO₂, which of these two compounds has a higher boiling point? Explain your answer.

.....
.....
.....
.....

[6]

[Total: 10]

- 3 (a) The initial rate of reaction between compound **A**, $\text{HCOOCH}_2\text{CH}_2\text{CH}_3$, and aqueous sodium hydroxide was measured in a series of experiments at a constant temperature. The data obtained was recorded and given below:

Expt.	Initial concentration of NaOH / mol dm^{-3}	Initial concentration of A / mol dm^{-3}	Initial rate / $\text{mol dm}^{-3} \text{ s}^{-1}$
1	0.040	0.030	4.0×10^{-4}
2	0.040	0.045	6.0×10^{-4}
3	0.060	0.045	9.0×10^{-4}
4	0.120	0.060	b

- (i) Use the data in the table to deduce the order of reaction with respect to **A** and the order of reaction with respect to NaOH.

- (ii) Hence write a rate equation for this experiment.

.....

- (iii) Calculate the initial rate of reaction, **b** of experiment 4.

[5]

- (b) In a further experiment at a different temperature, the initial rate of reaction was found to be $9.0 \times 10^{-3} \text{ mol dm}^{-3} \text{ s}^{-1}$ when the initial concentration of **A** was $0.020 \text{ mol dm}^{-3}$ and the initial concentration of NaOH was 2.00 mol dm^{-3} .

Under these new conditions, the reaction is first order with respect to **A** and appears to be zero order with respect to sodium hydroxide.

- (i) Write a rate equation for the reaction under these new conditions.

.....

- (ii) Calculate a value for the new rate constant, k' , under these new conditions and state its units.

[3]

(c) Compounds **P** and **Q** have the molecular formula C_3H_8O .

Both **P** and **Q** decolourises hot acidified $KMnO_4$ to form **R** and **S** respectively.

R reacts with Na_2CO_3 to produce an acidic gas but **S** does not.

S forms a crystalline orange precipitate with 2,4-dinitrophenylhydrazine but does not form a silver mirror with Tollen's reagent.

Draw one possible structure for **P**, **Q**, **R** and **S**, explaining your deductions clearly.

<u>Observations</u>	<u>Deductions</u>

Structure of P	Structure of Q
Structure of R	Structure of S

[7]

[Total: 15]

SECTION B (40 marks)

Answer only 2 out of the 3 questions

- 4 (a)** Ammonia is manufactured from nitrogen and hydrogen by the Haber Process.



- (i) Apart from its use in making fertilizers, give another reason why the manufacture of ammonia is important.
- (ii) Explain why the activation energy of the Haber Process is high.
- (iii) State Le Chatelier's Principle.
- (iv) Explain why the Haber Process is carried out under high pressure and low temperature conditions.
- (v) Write an expression for K_c , and hence calculate the value of K_c given the following equilibrium concentrations at 1000K. State the units of K_c .

	Concentration / mol dm ⁻³
N ₂	1.36
H ₂	1.84
NH ₃	0.142

[11]

- (b)** Consider the organic compound with the structural formula CH₃CH=CH(OH).

- (i) Define the term *structural isomers*.
- (ii) Draw **and** name the structures of two isomers of CH₃CH=CH(OH) that have **different** functional groups.

[4]

- (c)** Halogenoalkanes are useful in the laboratory as intermediates in making other organic compounds.

State in increasing order, the reactivity of iodoethane, bromoethane and chloroethane. Explain your answer.

[2]

- (d)** Butan-2-ol, CH₃CH₂CH(OH)CH₃, can undergo dehydration to form two products which are positional isomers of each other.

- (i) Suggest the reagents and conditions for the dehydration.
- (ii) Draw the structures of the two products.

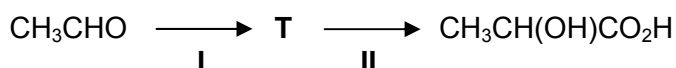
[3]

[Total: 20]

- 5 (a) Benzene can react with chlorine to form chlorobenzene.
- (i) What are the reagents and conditions needed for this reaction to occur?
 - (ii) Write a balanced equation for this reaction.
 - (iii) Explain why benzene undergoes substitution reactions but not addition reactions.

[3]

- (b) A sequence of reactions, starting from ethanal, is shown below.



- (i) Suggest the reagents and conditions for steps, I and II.
- (ii) Name the type of reaction for step I.
- (iii) Draw the structural formula of T.

[4]

- (c) Esters of carboxylic acids are often used as components in fruit flavouring. Ethyl butanoate, $\text{CH}_3\text{CH}_2\text{CH}_2\text{COOCH}_2\text{CH}_3$, is a major component of strawberry and pineapple flavouring.

What are the reagents and conditions needed to synthesise ethyl butanoate?

[3]

- (d) Potassium manganate is an oxidising agent commonly used in organic synthesis.

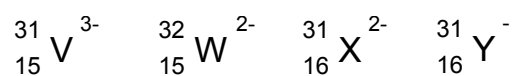
FA 1 contains Fe^{2+} ions. In an experiment, it was found that 25.0 cm^3 of **FA 1** required 18.00 cm^3 of $0.200 \text{ mol dm}^{-3}$ acidified KMnO_4 (aq) for complete reaction.

- (i) With reference to the *Data Booklet*, state the oxidation and reduction half-equations for this reaction.
- (ii) Hence, write an overall equation (with state symbols) for this reaction.
- (iii) Calculate the amount of Fe^{2+} present in 25.0 cm^3 of **FA 1**.

[5]

- (e) **Isoelectronic** atoms or ions have the same number of electrons. Those with the same number of neutrons are **isotonic**, and those with the same number of protons are **isotopic**.

- (i) Given the following ions,



Which of the following ions are:

- (I) Isoelectronic
 - (II) Isotonic
 - (III) Isotopic
- (ii) Which of the ion will be
- (I) deflected the most,
 - (II) deflected the least
- when passed through an electric field?

[5]

[Total: 20]

- 6 (a) Acetic acid is an organic compound with the chemical formula CH_3COOH . It is a colourless liquid that when undiluted is also called glacial acetic acid. It is a cigarette additive and, as the main component of vinegar, has a distinctive sour taste and pungent smell.

The acid dissociation constant K_a of ethanoic acid is $1.8 \times 10^{-5} \text{ mol dm}^{-3}$.

- (i) Write an expression for the K_a of ethanoic acid.
- (ii) Given that $[\text{H}^+] = \sqrt{[\text{weak acid}]K_a}$, calculate the $[\text{H}^+]$ and the pH of a 0.10 mol dm^{-3} solution of ethanoic acid.
- (iii) Explain why ethanoic acid is a weak acid.

[4]

- (b) A solution containing ethanoic acid and its salt, sodium ethanoate, $\text{CH}_3\text{CO}_2\text{Na}$, acts as a buffer solution.

- (i) What does the term *buffer solution* mean?
- (ii) Describe with equations how a buffer solution works when a small amount of H^+ ions and OH^- ions are added to it.

[3]

- (c) The relative molecular mass of ethanoic acid is 60. However, when ethanoic acid is dissolved in benzene, it is found that the relative molecular mass of the acid is 120. Explain this phenomenon using a clearly labeled diagram.

[2]

- (d) The standard enthalpy of neutralisation can be determined when solutions of an acid and a base are mixed. The container used for measuring the change in temperature is called a calorimeter.

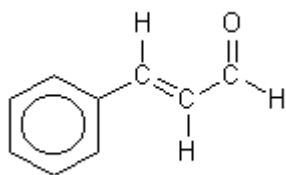
- (i) 250 cm^3 of sodium hydroxide solution of concentration $0.400 \text{ mol dm}^{-3}$ was added to 250 cm^3 of ethanoic acid of concentration $0.400 \text{ mol dm}^{-3}$ in a calorimeter. The initial temperature of the two solutions was 17.05°C . At the end of the reaction, the temperature rose to 19.55°C .

Assuming the specific heat capacity of all the solutions is $4.2 \text{ J g}^{-1} \text{ K}^{-1}$, calculate the standard enthalpy of neutralisation **per mole of water formed**.

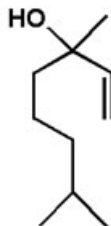
- (ii) State one possible source of error in this experiment.

[5]

- (e) The following compounds are two of the many chemicals present in cinnamon.



A



B

- (i) Describe **two** chemical tests that can be carried out to distinguish between both compounds. You should also state observations for each of the tests in your answers.
- (ii) Describe one chemical test that can give the same positive result when carried out on both compounds. State the observations that you will see.

[6]

[Total: 20]

End of Paper